

AWS checklist and maintenance guide

Station name	TAS-U	
People present	HCIT, MOLA	instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	11:50 on 3 AUG 2014	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from:	
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %:	

Notes:

station partially in reverse

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	AWS LAGS 6 sec +/-
Adjust logger time to UTC (be sure PC runs on UTC time)	Y/N
Download data to PC and/or change CF Card (wait for green light)	D/L + copy PC Y/N
Name of logger program	PROHLC 2013C
Photo or screen dump of values in fast scan mode	Y/N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	-73.1 °
Mast tilt across boom looking from radiometer (- if clockwise)	78.5 °
Boom direction relative to north	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.	Y/N
Radiometer aligned with mast and inclinometer? If not, measure.	Y/N
Temperature / humidity height (bottom of casing – surface)	~200 cm
Sonic ranger on AWS height (membrane – surface)	~210 cm
Wind sensor height (center of propeller – surface)	~240 cm
Sonic ranger on stakes height (membrane – surface)	~305 cm
Free length of stake – outer, holding boom with sonic ranger	325 cm
Free length of stake – middle	327 cm
Free length of stake – outer, holding boom without sonic ranger	409 cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	~1700 OUT cm
Height of ablation hose fluid level above the ice surface	50 * cm
Vertical difference in surface height (at station – at hose) (approx)	0 cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	120 cm
Stake assembly still standing?	Y/N / more or less
Apparent damage to sensors? If yes, which?	PRESSURE HOSE, BLADDER, BLOWN, BLADDER HANGING

Notes:

Stakes on ice, ~~some~~ shredding on side of crevasse
 a crevasse
 * Bladder had fallen off of front of the damaged box and was hanging by the propeller and the sensor cable. Fluid column interrupted however, and highest point of it ~45 cm above surface

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer	080164	=	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer	00089		- Aligned with radiometer within 0.5°: Y/N
Satellite antenna	GEUS 12		
Wind sensor	110446	=	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly	6/2011	=	
T/hum assembly casing			- Fan spins and sounds OK: ☒ Y/N
T/hum assembly bracket	06/2011		
Temperature probe in T/hum assembly	HP100	60179402 =	
HygroClip in T/hum assembly	51382185	5502517	- 40 °C offset in HygroClip temperature: Y/ ☒ N
Sonic ranger on AWS	1736	5111	- Old sensor with new membrane: Y/ ☒ N - Clicking 5 times as it should: Y/N
Solar panel	UNREPAIRABLE	=	- Output OK: ☒ Y/N - Clean: ☒ N
GPS antenna	036110634	=	- Stuck to top of logger enclosure: ☒ Y/N
Data logger	E3201	=	- New internal battery in old logger: Y/ ☒ N - New operating system: Y/ ☒ N
Card reader	4442	=	- LED green of flashing orange (active): ☒ Y/N
Multiplexer	E4796	=	- Clicking as it should: ☒ Y/N
Iridium modem IMEI number	1218580	=	- Transmissions tested: ☒ Y/N - Problems: Y/ ☒ N
Barometer	38140P2	=	
Sonic ranger on stakes	UNREPAIRABLE	5935	- Old sensor with new membrane: Y/ ☒ N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: ☒ Y/N
Thermistor string	R31	B21	- Old string left in ice: ☒ Y/N/cut
Ablation hose assembly	UNREPAIRABLE	=	- Old hose left in ice: ☒ Y/N/cut - Air in hose: Y/ ☒ N/removed - Bladder 50% full: ☒ Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): ☒ Y/N - New vent filter in logger enclosure: Y/ ☒ N - Moisture inside: Y/ ☒ N - New desiccant bags (2): Y/N		
Battery box	- New box with new batteries: Y/ ☒ N - Old box with new batteries: Y/ ☒ N - Moisture inside: Y/N <i>not opened</i> - New desiccant bags (2): Y/ ☒ N		
Stakes	- New ones drilled in: ☒ Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: ☒ Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	(Y) N
Name of logger program	--2014d
Photo or screen dump of values in fast scan mode	(Y) N
Latitude (dd mm.mmm)	Same
Longitude (dd mm.mmm)	Same
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	-4.9 °
Mast tilt across boom looking from radiometer (- if clockwise)	-1.5 °
Boom direction relative to north (radiometer should point to true south)	58° true/magn
Box on wind sensor exactly towards mast? If not, change.	(Y) N
Radiometer aligned with mast and inclinometer? If not, align.	(Y) N
Temperature / humidity height (bottom of casing – surface)	281 cm
Sonic ranger on AWS height (membrane – surface)	297 cm
Wind sensor height (center of propeller – surface)	340 cm
Sonic ranger on stakes height (membrane – surface)	98 cm
Free length of stake – outer, holding boom with sonic ranger	121 cm
Free length of stake – middle	116 cm
Free length of stake – outer, holding boom without sonic ranger	130 cm
Free length of ablation hose on/above the ice surface	20.8 m cm
Alt: depth of ablation hose using markings on hose	— cm
Height of ablation hose fluid level above the ice surface	~190 cm
Vertical difference in surface height (at station – at hose) (approx)	0 cm
Length of thermistor string on surface from surface marking	32 cm
Height of surface irregularities	100 cm

Notes:

Station placed slightly elevated

Terrain difficult !!

Very bumpy

Brushed 1440 LCL

Page 4 of 4 > 9 stakes (2m) + 1 boom on site

